

LLM audit-logic hop

Detecting a sudden change in the reviewer map $P(Y | X)$

OnlineRFPerm landing note (RFPerm-as-Judge)

2026-09-16

Abstract

The LLM **audit / moderation / preference-judge** stream is the alignment counterpart of hallucination regime detection. Here Y is the reviewer decision (pass/reject, or chosen/rejected), not a factuality bit. Last-two OnlineRFPerm asks whether the **audit map** hopped between adjacent windows. A slowly stricter policy can remain smooth. A overnight judge swap, policy-pack cut, or annotator-pool replacement is the changepoint—“the reviewer died suddenly”, not “the model forgot how to write.” We post the HH-RLHF (Bai et al., 2022) RFPerm-as-Judge numbers already shipped: quiet $t = 2$ silent, cut $t = 4$ fires with delay 0, judge error $0.147 \rightarrow 0.539$.

1 What is being tested

Each batch is (X_t, Y_t) : content to review, and the current auditor’s label. The previous-window probe μ_0 is yesterday’s audit logic. The hop gate is consecutive OOS, $\gamma = 1.25$:

$$\text{fire}_t = \mathbf{1}\{e_{\text{now}}/e_{\text{prev}} \geq \gamma\}.$$

High reject rate is not a hop. A stable-strict auditor is still smooth. The peculiar jump is when adjacent maps stop looking like one another.

Table 1: Two layers for LLM audit. Smoothness is consecutive-window changepoint; trustworthiness is degradation versus a frozen launch probe. A slowly stricter policy can stay smooth; a overnight policy/judge swap is the hop.

Question	Statistic	Audit reading
Is the audit chain still smooth?	Last-two hop: $e_{\text{now}}/e_{\text{prev}} \geq \gamma$	Same rubric, same judge: do adjacent batches still look like one map?
Is the auditor still trustworthy?	RFPerm vs frozen launch probe	Even a silky tightening can be caught before the old gold label is unusable.

Table 2: RFPerm-as-Judge on Anthropic HH-RLHF helpful-base ($n = 4000$, $50 \text{ batches} \times 80$, cut B_4 , $\gamma = 1.25$). Quiet $t = 2$ does not fire; the cut hop fires in the same window (delay 0). Judge error $0.147 \rightarrow 0.539$ is the regime, not “the judge suddenly became stupid.”

Window	Fired	Ratio	Judge err / mix	Reading
Quiet $t = 2$	no	1.03	pos-rate 0.48	audit map still smooth
Cut $t = 4$	yes	1.34	pos-rate 0.59	preference map hopped
Post-cut regime	—	3.67	$0.147 \rightarrow 0.539$	new audit law, then can be smooth again

Table 3: What moved on the HH-RLHF cut. Style-domain AUC flags a $P(X)$ register shift; last-two fire + judge-err ratio flags $P(Y | X)$. Text-domain AUC near 1/2 says this was not a bag-of-words topic swap.

Axis	Number	Action if this axis dominates
Audit map $P(Y X)$	hop ratio 1.34; err ratio 3.67	do not treat current judge as gold; Top- k <code>po_risk0</code> re-review; block
Register $P(X)$	style AUC 0.999	refresh the portrait / routing features, not necessarily the rubric
Lexical mix	text AUC 0.494	no action from bag-of-words domain shift

Table 4: Serving gate. Fire opens a contrast window; it is not a violation classifier. `po_risk0` only ranks rows after a hop.

State	Meaning	Action
quiet	consecutive audit logic holds	standard path; merge prefs under the old gate; $w = 1$
fire	audit logic hopped	current judge is not gold; human Top- k ; refuse or cap DPO merge; optional $\sqrt{\text{PO}}$ on $T = 1$ only

2 Two layers

Smoothness (last-two): did the rubric break between $t - 1$ and t ? **Trustworthiness** (frozen launch probe): even a silky week-by-week tightening can be rejected versus the original gold before the auditor is unusable. Keeping only last-two would miss the boiling-frog. Keeping only the launch probe would call every slow tightening a crash. The robust serving rule uses both.

3 What fire is allowed to mean

On fire: the current judge is not gold; open a human contrast window on Top- k `po_risk0`; cap or refuse DPO/RLHF merges from that window. On quiet: standard path, $w = 1$. `po_risk0` is residual strength versus the previous probe—not a violation score, not a moral verdict, not unique causal attribution.

References

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